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Time Series Analysis

Statistics · Decomposition · Stationarity · Non-stationarity · ARMA ·
ARCH / GARCH

Lecture notes

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Time Series Analysis — Lecture notes.

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Foreword

These lecture notes accompany the *Time Series Analysis* course taught at Esprit School of Business. They gather, in a single document, six chapters that build progressively the modern toolbox of applied time-series analysis.

Chapter 1 reviews the basic statistical concepts (moments, estimation, sampling distributions, inference) needed throughout the course. It serves both as a refresher and as the conceptual reference for the chapters that follow.

Chapter 2 presents the *classical decomposition* of a time series into trend, seasonality, and residual. Trend estimation by polynomial fitting and centred moving averages, and seasonality estimation by seasonal averaging or Fourier coefficients, are derived explicitly.

Chapter 3 introduces *strict* and *weak* stationarity, the autocovariance and partial autocorrelation functions, the Wold decomposition, and the Box–Jenkins reading of the sample ACF/PACF. The Ljung–Box test for white noise is stated and proved at the level required for the labs.

Chapter 4 addresses *non-stationary* series. The backshift B and difference Δ operators are introduced, integration order $I(d)$ is defined, and the unit-root tests — Augmented Dickey–Fuller (ADF), KPSS, Phillips–Perron — are presented in detail. $\text{ARIMA}(p, d, q)$ and $\text{SARIMA}(p, d, q)(P, D, Q)_s$ models close the chapter.

Chapter 5 is the central chapter on ARMA models. $\text{AR}(p)$, $\text{MA}(q)$ and $\text{ARMA}(p, q)$ are defined in lag-operator form; stationarity and invertibility are stated as polynomial-root conditions; and the full Box–Jenkins workflow — identification, MLE estimation, AIC/BIC selection, residual diagnostics, forecasting — is laid out.

Chapter 6 closes the module with $\text{ARCH}(q)$ and $\text{GARCH}(p, q)$ models for conditional heteroskedasticity. Volatility clustering, Engle’s LM test, quasi-MLE estimation, and one-day Value-at-Risk are presented; the chapter mentions the EGARCH and GJR-GARCH extensions for the leverage effect.

Each chapter is delivered with a Beamer Metropolis 16:9 slide deck, a Python lab (statement, solution, student skeleton, reference code), and a project consolidating the chapter’s notions. A final integrative project, a reflective activity, and the official ECUE module sheet (in French) complete the module.

Happy reading.

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Chapter 1

Basic statistical concepts

Introduction

Before studying time series proper, this chapter reviews the statistical concepts on which the rest of the course relies: *moments* of a distribution, classical *point estimators*, *Method of Moments* and *Maximum Likelihood Estimation*, sampling distributions, and basic *inference* (confidence intervals and hypothesis tests).

The convention throughout the chapter is $X_1, \dots, X_n$ are *independent and identically distributed* (i.i.d.) draws from a distribution with density $f(x; \theta)$ parameterised by some unknown θ . We observe a sample and want to learn θ . Time series will later violate the i.i.d. assumption, but the basic estimators introduced here remain the building blocks.

1.1 Moments of a distribution

Definition 1 – Raw and central moments

Let X be a real-valued random variable with density f and mean $\mu = \mathbb{E}[X]$.

- The *raw ℓ -th moment* is $m'_\ell = \mathbb{E}[X^\ell] = \int x^\ell f(x) dx$.
- The *central ℓ -th moment* is $m_\ell = \mathbb{E}[(X - \mu)^\ell]$.

Useful quantities.

$$\begin{aligned} \mu &= m'_1, & \sigma^2 &= m_2 = \text{Var}[X], \\ \text{skewness } S(X) &= \frac{m_3}{\sigma^3}, & \text{kurtosis } K(X) &= \frac{m_4}{\sigma^4}. \end{aligned}$$

Remark – Interpretation

- $S(X)$ measures *asymmetry*. $S = 0$ for any symmetric distribution; $S > 0$ for a right-skewed distribution.
- $K(X)$ measures *tail-heaviness*. The Gaussian has $K = 3$; the *excess kurtosis* $K - 3$ is therefore the standard comparison: $K - 3 > 0$ indicates heavier tails than the Gaussian (leptokurtic).

Example 2 – Moments of the standard normal

For $X \sim \mathcal{N}(\mu, \sigma^2)$ one shows by direct integration or via the moment-generating function:

$$\mathbb{E}[X] = \mu, \quad \text{Var}[X] = \sigma^2, \quad m_3 = 0, \quad m_4 = 3\sigma^4.$$

Hence $S(X) = 0$ and $K(X) = 3$.

1.2 Classical estimators and sampling distributions

1.2.1 Sample mean and sample variance

Definition 3 – Empirical estimators

Given i.i.d. data $X_1, \dots, X_n$:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Proposition 4 – Unbiasedness of $\bar{X}_n$ and S_n^2

For i.i.d. data with finite second moments, $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathbb{E}[S_n^2] = \sigma^2$.

Proof. For the mean: $\mathbb{E}[\bar{X}_n] = \frac{1}{n} \sum_i \mathbb{E}[X_i] = \mu$ by linearity. For the variance, expand:

$$\sum_i (X_i - \bar{X}_n)^2 = \sum_i X_i^2 - n \bar{X}_n^2.$$

Take expectations: $\mathbb{E}[X_i^2] = \mu^2 + \sigma^2$ and $\mathbb{E}[\bar{X}_n^2] = \mu^2 + \sigma^2/n$. Hence $\mathbb{E}[\sum_i (X_i - \bar{X}_n)^2] = n(\mu^2 + \sigma^2) - n(\mu^2 + \sigma^2/n) = (n-1)\sigma^2$. The $n-1$ denominator in S_n^2 ensures $\mathbb{E}[S_n^2] = \sigma^2$. $\square$

Remark – Why $n-1$?

Replacing the unknown μ by $\bar{X}_n$ in the variance formula “costs” one degree of freedom. Dividing by n would systematically underestimate σ^2 .

1.2.2 Sampling distribution under normality

Theorem 5 – Cochran-type results

If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then:

1. $\bar{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$;
2. $\frac{(n-1)S_n^2}{\sigma^2} \sim \chi_{n-1}^2$;
3. $\bar{X}_n$ and S_n^2 are independent;
4. $\frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \sim t_{n-1}$.

Remark

Result (4) follows from (1)–(3): the numerator is standard normal, the denominator is the square root of an independent χ_{n-1}^2 divided by its degrees of freedom — the very definition of Student’s t .

1.2.3 Asymptotic results

Theorem 6 – Law of Large Numbers

For i.i.d. data with $\mathbb{E}[X_i] = \mu$:

$$\bar{X}_n \xrightarrow{p} \mu \quad \text{as } n \rightarrow \infty.$$

Theorem 7 – Central Limit Theorem

For i.i.d. data with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

Remark – Practical consequence

Even when the X_i are not Gaussian, the sample mean behaves approximately as a normal random variable for large n . The rule of thumb $n \geq 30$ is conservative; in practice the convergence is faster for symmetric, light-tailed distributions and slower for skewed or heavy-tailed ones.

1.3 Estimation methods

1.3.1 Method of Moments (MoM)

Definition 8 – Method of Moments

Let $\theta \in \mathbb{R}^k$. The MoM estimator $\hat{\theta}_n^{\text{MoM}}$ solves the system of equations

$$\mathbb{E}_\theta[X^p] = \frac{1}{n} \sum_{i=1}^n X_i^p, \quad p = 1, 2, \dots, k.$$

That is, theoretical moments are equated to sample moments.

Example 9 – $\mathcal{N}(\mu, \sigma^2)$ via MoM

With $\theta = (\mu, \sigma^2)$ and $k = 2$:

$$\begin{aligned} \mu &= \bar{X}_n, \\ \mu^2 + \sigma^2 &= \frac{1}{n} \sum_i X_i^2. \end{aligned}$$

Subtracting yields $\hat{\sigma}_{\text{MoM}}^2 = \frac{1}{n} \sum (X_i - \bar{X}_n)^2$, which differs from S_n^2 by a factor $n/(n-1)$.

1.3.2 Maximum Likelihood Estimation (MLE)

Definition 10 – Likelihood and MLE

For i.i.d. data with density $f(x; \theta)$:

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta), \quad \ell(\theta) = \log L(\theta).$$

The MLE is $\hat{\theta}_n^{\text{MLE}} = \arg \max_\theta \ell(\theta)$.

Example 11 – $\mathcal{N}(\mu, \sigma^2)$ via MLE

The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2.$$

Setting $\partial \ell / \partial \mu = 0$ gives $\hat{\mu}_{\text{MLE}} = \bar{X}_n$. Then setting $\partial \ell / \partial \sigma^2 = 0$ gives $\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum (X_i - \bar{X}_n)^2$. So MLE coincides with MoM here, and both are biased for σ^2 (their expectation is $\frac{n-1}{n} \sigma^2$).

Theorem 12 – Asymptotic normality of the MLE

Under regularity conditions (identifiability, smoothness, finite Fisher information $I(\theta_0) > 0$):

$$\sqrt{n} (\hat{\theta}_n^{\text{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}).$$

Remark – MoM vs. MLE

- MoM is computationally simple — just match moments.
- MLE is *asymptotically efficient*: its variance attains the Cramér–Rao lower bound $I(\theta_0)^{-1}/n$.
- Both estimators are consistent (under regularity).
- For “nice” families (Gaussian, exponential, Poisson), MoM and MLE often coincide or are very close.

1.4 Multivariate data

1.4.1 Joint, marginal, and conditional distributions

Definition 13 – Joint and marginal density

For two random variables X, Y with joint density $f(x, y)$:

$$f_X(x) = \int f(x, y) dy, \quad f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)}.$$

The chain rule reads $f(x, y) = f_{Y|X}(y | x) f_X(x)$.

Definition 14 – Independence

X and Y are *independent* iff $f(x, y) = f_X(x) f_Y(y)$ for all x, y .

1.4.2 Covariance and correlation

Definition 15 – Covariance and correlation

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)], \quad \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}X \text{Var}Y}}.$$

Proposition 16 – Sample versions

$$\widehat{\text{Cov}}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}), \quad \widehat{\text{Corr}}(X, Y) = \frac{\widehat{\text{Cov}}(X, Y)}{\hat{\sigma}_X \hat{\sigma}_Y}.$$

The sample correlation lies in $[-1, 1]$.

Warning – Correlation captures only linear dependence

$\text{Corr}(X, Y) = 0$ does not imply independence. Counter-example: let X be symmetric around 0 and $Y = X^2$. Then $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0 = \mathbb{E}[X]\mathbb{E}[Y]$, hence $\text{Cov}(X, Y) = 0$, yet Y is a deterministic function of X .

1.5 Statistical inference

1.5.1 Confidence intervals for the mean

Definition 17 – Confidence interval

A $(1 - \alpha)$ -confidence interval for μ is a random interval $[\hat{L}_n, \hat{U}_n]$ such that $\mathbb{P}(\hat{L}_n \leq \mu \leq \hat{U}_n) = 1 - \alpha$ in repeated sampling.

Gaussian sample, known σ (z -interval). Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ known. By Theorem 1.2.2 (1):

$$\text{CI}_{1-\alpha}(\mu) = \left[\bar{X}_n - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

Gaussian sample, unknown σ (t -interval). Replace σ by S_n and use Student's t (Theorem 1.2.2 (4)):

$$\text{CI}_{1-\alpha}(\mu) = \left[\bar{X}_n - t_{n-1, 1-\alpha/2} \frac{S_n}{\sqrt{n}}, \bar{X}_n + t_{n-1, 1-\alpha/2} \frac{S_n}{\sqrt{n}} \right].$$

1.5.2 Hypothesis testing

Definition 18 – Test on the mean

Two-sided test: $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$. Test statistic

$$T = \frac{\bar{X}_n - \mu_0}{S_n / \sqrt{n}} \sim t_{n-1} \text{ under } H_0.$$

Reject H_0 at level α if $|T| > t_{n-1, 1-\alpha/2}$, i.e. the p -value $2 \cdot \mathbb{P}(T_{n-1} > |T|) < \alpha$.

Remark – Errors

- Type I error (size of the test): $\alpha = \mathbb{P}(\text{reject } H_0 \mid H_0)$.
- Type II error: $\beta = \mathbb{P}(\text{not reject } H_0 \mid H_1)$.
- *Power* of the test: $1 - \beta$, the probability of correctly rejecting H_0 when H_1 is true.

Example 19 – Numerical illustration

Suppose $n = 50$ Atlanta monthly average temperatures with sample mean $\bar{X}_{50} = 61.2$ and sample standard deviation $S_{50} = 16.8$. Test $H_0 : \mu = 60$ against $H_1 : \mu \neq 60$ at $\alpha = 0.05$.

$$T = \frac{61.2 - 60}{16.8 / \sqrt{50}} \approx 0.505, \quad t_{49, 0.975} \approx 2.010.$$

Since $|T| < 2.010$, we do *not* reject H_0 ; p -value $\approx 2 \cdot \mathbb{P}(t_{49} > 0.505) \approx 0.616$. The corresponding 95% CI is $[61.2 - 4.78, 61.2 + 4.78] = [56.4, 66.0]$, which contains $\mu_0 = 60$, consistent with the test conclusion.

Warning – Common pitfalls

- “Failing to reject H_0 ” is *not* “proving H_0 ”.
- A small p -value does not imply a large effect — it only signals statistical significance, not practical importance.
- Multiple testing inflates Type I error: use Bonferroni or FDR corrections when running many tests.

Summary

Quantity	Estimator	Sampling distribution (Gaussian data)
μ	$\bar{X}_n = \frac{1}{n} \sum X_i$	$\mathcal{N}(\mu, \sigma^2/n)$
σ^2	$S_n^2 = \frac{1}{n-1} \sum (X_i - \bar{X}_n)^2$	$(n-1)S_n^2/\sigma^2 \sim \chi_{n-1}^2$
Pivot	$T = (\bar{X}_n - \mu)/(S_n/\sqrt{n})$	t_{n-1}

What to remember.

- Two estimation routes: MoM (match moments) and MLE (maximise likelihood). Both consistent; MLE asymptotically efficient.
- Under normality, the sampling distribution of $\bar{X}$ is exactly Gaussian; the CLT extends this asymptotically.
- Confidence intervals and hypothesis tests on the mean rely on Student's t when σ is unknown.

Chapter 2

Time series decomposition

Introduction

The first step in time-series analysis is to *decompose* the observed data into structural components: a slowly varying *trend* T_t , a periodic *seasonality* S_t , and a stationary residual ε_t . The trend and seasonality carry the deterministic structure; the residual captures the stochastic fluctuations and is the input to the modelling techniques of Chapters 3–6.

This chapter presents:

1. the additive and multiplicative decomposition models;
2. three trend-estimation strategies (moving average, parametric polynomial regression, non-parametric LOESS / splines);
3. three seasonality-estimation strategies (seasonal averaging, dummy regression, cosine–sine Fourier models);
4. residual diagnostics and the full decomposition workflow.

2.1 The decomposition model

Definition 1 – Additive vs. multiplicative model

For an observed time series $(Y_t)_{t=1}^T$ with seasonal period d :

$$\text{Additive: } Y_t = T_t + S_t + \varepsilon_t, \quad \text{Multiplicative: } Y_t = T_t \cdot S_t \cdot \varepsilon_t.$$

Here T_t is a slowly varying *trend*, S_t is a periodic *seasonal component* of period d (i.e. $S_{t+d} = S_t$), and ε_t is a zero-mean stationary residual (additive case) or a positive multiplicative perturbation with $\mathbb{E}[\varepsilon_t] = 1$ (multiplicative case).

Remark

The two models differ qualitatively:

- In the additive model, the seasonal swing has a *constant amplitude* regardless of the level of the series.
- In the multiplicative model, the swing *grows proportionally* to T_t .
- Taking logarithms converts a multiplicative model into an additive one: $\log Y_t = \log T_t + \log S_t + \log \varepsilon_t$.

This log transform is standard for positive series with growing amplitude (sales, population, GDP).

Identifiability. Without further constraints the decomposition is not unique (a constant can move from T to S). Two standard normalisations:

- $\sum_{k=1}^d S_k = 0$ in the additive case (seasonality sums to zero over a full cycle);
- $\sum_{k=1}^d S_k = d$ in the multiplicative case (seasonality averages to one).

2.2 Trend estimation

2.2.1 Symmetric (centred) moving average

Definition 2 – Centred moving average

For an odd window $2k + 1$ with $k \in \mathbb{N}$:

$$\hat{T}_t = \frac{1}{2k+1} \sum_{i=-k}^k Y_{t+i}, \quad t = k+1, \dots, T-k.$$

Boundary points $t = 1, \dots, k$ and $t = T - k + 1, \dots, T$ are not estimated.

Proposition 3 – The centred MA preserves linear trends

If $Y_t = a + bt$ exactly, then $\hat{T}_t = a + bt$ for all interior t .

Proof. Substitute $Y_{t+i} = a + b(t+i)$:

$$\hat{T}_t = \frac{1}{2k+1} \sum_{i=-k}^k (a + b(t+i)) = a + bt + \frac{b}{2k+1} \underbrace{\sum_{i=-k}^k i}_{=0} = a + bt.$$

□

Proposition 4 – The centred MA cancels period- $(2k+1)$ seasonality

If S_t has period $d = 2k + 1$ and $\sum_{k=1}^d S_k = 0$, then $\sum_{i=-k}^k S_{t+i} = 0$ for all t , hence the MA filters out S_t exactly.

Remark

For *even* period d (e.g. $d = 12$ for monthly data), one uses a $2 \times d$ centred MA: $\hat{T}_t = \frac{1}{2d} (\frac{1}{2}Y_{t-d/2} + Y_{t-d/2+1} + \dots + Y_{t+d/2-1} + \frac{1}{2}Y_{t+d/2})$. This combines two consecutive symmetric MAs of length d to yield a window of length $d + 1$ with end-points down-weighted.

Example 5 – Numerical illustration

$Y = (10, 12, 15, 14, 16, 18, 20)$, $k = 1$ (window 3):

$$\hat{T}_2 = \frac{37}{3} \approx 12.33, \quad \hat{T}_3 = \frac{41}{3} \approx 13.67, \quad \hat{T}_4 = \frac{45}{3} = 15.00,$$

$$\hat{T}_5 = 16.00, \quad \hat{T}_6 = 18.00.$$

The estimated trend follows the upward drift of the series; $t = 1$ and $t = 7$ are lost.

2.2.2 Parametric polynomial regression

Definition 6 – Polynomial trend

$T_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \dots + \beta_p t^p$ estimated by ordinary least squares (OLS) on (t, Y_t) .

Remark

- $p = 1$: straight line. Use only when the eyeball plot is consistent with a linear pattern.
- $p = 2, 3$: flexible enough for most economic / climate data on a moderate horizon.
- $p \geq 4$: usually overfit and behave poorly outside the sample range. Prefer LOESS or splines.

Use AIC / BIC, cross-validation or out-of-sample MSE to select p .

2.2.3 Non-parametric regression**Definition 7 – LOESS / local polynomial regression**

At each t fit a low-degree polynomial on a *neighbourhood* of t , with weights decreasing in $|t' - t|$ (e.g. tricube kernel). The fraction of data used in each local fit is the *span* parameter $\alpha \in (0, 1)$.

Why non-parametric?

- Captures arbitrary smooth trends without imposing a global functional form.
- Robust variants (`loess(..., family="symmetric")`) are less affected by outliers.
- Span α small $\Rightarrow$ wiggly fit; large $\Rightarrow$ over-smooth. Default $\alpha = 0.75$ is often too aggressive for time-series trend extraction; $\alpha = 0.1$ – 0.3 is usually preferred.

Remark – Other smoothers

Cubic smoothing splines minimise $\sum (Y_t - g(t))^2 + \lambda \int g''(s)^2 ds$ over twice-differentiable g . The penalty λ trades off fidelity vs. smoothness; $\lambda \rightarrow 0$ recovers interpolation, $\lambda \rightarrow \infty$ a linear fit. Wavelets offer multi-resolution alternatives but are seldom needed for a one-step decomposition.

2.3 Seasonality estimation**2.3.1 Seasonal averaging****Definition 8 – Seasonal averaging after detrending**

Let $\tilde{Y}_t = Y_t - \hat{T}_t$ be the detrended series. For each season $k = 1, \dots, d$ define

$$\hat{S}_k = \frac{1}{n_k} \sum_{t: \text{season}(t)=k} \tilde{Y}_t,$$

where n_k is the number of observations falling in season k . Re-centre by subtracting the overall mean to ensure $\sum_k \hat{S}_k = 0$.

Example 9 – Monthly seasonal index

With monthly data ($d = 12$) over 10 years, $n_k = 10$ for each k . The estimate $\hat{S}_{\text{July}}$ is the average of the ten July deviations from the trend.

2.3.2 Dummy-variable regression**Definition 10 – Seasonal dummies**

Let $D_{k,t} = \mathbf{1}_{\{\text{season}(t)=k\}}$ for $k = 2, \dots, d$. Estimate by OLS:

$$Y_t = \beta_0 + \sum_{k=2}^d \beta_k D_{k,t} + \varepsilon_t.$$

The omitted season ($k = 1$) is the *baseline*; β_k measures the seasonal deviation of season k from the baseline.

Remark – Why $d - 1$ dummies?

Including all d dummies plus an intercept produces perfect multicollinearity ($\sum_k D_{k,t} = 1$ for all t). Drop one season or drop the intercept.

Remark – Combining trend and seasonality

A standard “one-shot” specification estimates trend and seasonality *jointly*:

$$Y_t = \beta_0 + \beta_1 t + \sum_{k=2}^d \gamma_k D_{k,t} + \varepsilon_t.$$

This avoids the residual contamination that arises when one estimates the trend first by MA then the seasonality from the detrended series.

2.3.3 Cosine–sine (Fourier) model

Definition 11 – Trigonometric seasonality

With angular frequency $\omega_d = 2\pi/d$:

$$S_t = \sum_{j=1}^J [a_j \cos(j \omega_d t) + b_j \sin(j \omega_d t)], \quad J \leq \lfloor d/2 \rfloor.$$

The coefficients (a_j, b_j) are estimated by OLS in the regression $Y_t = \beta_0 + S_t + \varepsilon_t$ (or jointly with a trend).

Proposition 12 – Equivalence with dummies

With $J = \lfloor d/2 \rfloor$ harmonics (and the constant), the trigonometric basis spans the same subspace as the seasonal dummies, hence yields the same fitted seasonal pattern.

Remark

The Fourier model is *parsimonious*: with $J = 1$ harmonic only, seasonal swings are described by 2 parameters instead of $d - 1$. This helps when d is large (e.g. $d = 365$ for daily data with annual seasonality).

2.4 Residuals and diagnostics

After decomposition, the residual is

$$\hat{\varepsilon}_t = Y_t - \hat{T}_t - \hat{S}_t.$$

Remark – What to check

- **No remaining trend.** Plot $\hat{\varepsilon}_t$ vs. t ; any systematic drift means under-fit trend.
- **No remaining seasonality.** Compute the autocorrelation function (Chapter 3) and look for spikes at lag d (and its multiples).
- **Constant variance (homoscedasticity).** A funnel-shaped plot signals heteroscedasticity, suggesting a multiplicative rather than additive model — or ARCH/GARCH effects (Chapter 6).

- **Approximate normality.** Histogram + QQ-plot. Important for the validity of t -based confidence intervals.

Warning

A common mistake is to declare the decomposition successful as soon as the residual “looks small”. The residual must be *structureless* — that is, free of autocorrelation, of remaining seasonality and of heteroscedasticity. The next chapters provide the formal tools to check this.

2.5 Decomposition workflow

The full pipeline:

1. **Plot** Y_t versus t . Decide additive vs. multiplicative from the visual amplitude pattern.
2. If multiplicative, apply the log transform.
3. **Estimate the trend** $\hat{T}_t$:
 - moving average (centred, $2 \times d$ for even d);
 - polynomial regression ($p = 1, 2$ or 3);
 - LOESS / spline.
4. **Detrend:** $\tilde{Y}_t = Y_t - \hat{T}_t$.
5. **Estimate the seasonality** $\hat{S}_t$ from $\tilde{Y}_t$:
 - seasonal averaging (re-centred);
 - dummy regression;
 - cosine–sine model.
6. **Residual:** $\hat{\varepsilon}_t = Y_t - \hat{T}_t - \hat{S}_t$.
7. **Diagnostics** on $\hat{\varepsilon}_t$ (Section 2.4).
8. If diagnostics pass: model the residual with ARMA (Chapter 5).

Summary

Component	Method	Pros / Cons
Trend	Moving average ($2k + 1$)	simple, loses k ends
	Polynomial regression	global, risk of overfit
	LOESS / splines	flexible, span tuning
Seasonality	Seasonal averaging	simple, no model
	Dummy regression	joint with trend
	Cosine–sine (Fourier)	parsimonious, multi-harmonic

What to remember.

- Decomposition is the *first step* of any time-series analysis; the rest of the course works on the residual.
- Choose additive or multiplicative based on the amplitude pattern of the seasonality.
- Estimate trend and seasonality jointly (dummy or Fourier regression) when feasible — it controls residual contamination.

Chapter 3

Stationary time series

Introduction

After decomposing the observed series into trend, seasonality and residual (Chapter 2), we usually want the residual to be *stationary* so that the modelling tools of Chapter 5 apply. This chapter formalises stationarity, introduces the autocovariance and autocorrelation functions (ACF and PACF), states the Wold decomposition theorem, and provides *white-noise tests* (Box–Pierce, Ljung–Box) to validate model residuals.

3.1 Stationarity

3.1.1 Strict and weak stationarity

Definition 1 – Strict (strong) stationarity

A process $\{X_t\}_{t \in \mathbb{Z}}$ is *strictly stationary* if, for every integer k , every $t_1 < \dots < t_k$ and every $h \in \mathbb{Z}$,

$$(X_{t_1}, X_{t_2}, \dots, X_{t_k}) \stackrel{d}{=} (X_{t_1+h}, X_{t_2+h}, \dots, X_{t_k+h}).$$

Definition 2 – Weak (covariance) stationarity

A process $\{X_t\}$ with $\mathbb{E}[X_t^2] < \infty$ is *weakly stationary* if there exist $\mu \in \mathbb{R}$ and a function $\gamma : \mathbb{Z} \rightarrow \mathbb{R}$ such that

$$\mathbb{E}[X_t] = \mu, \quad \text{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{for all } t, h \in \mathbb{Z}.$$

The function γ is the *autocovariance function* (ACVF).

Proposition 3 – Strict $\Rightarrow$ weak (under L^2)

If $\{X_t\}$ is strictly stationary with $\mathbb{E}[X_t^2] < \infty$, then $\{X_t\}$ is weakly stationary.

Proof. By strict stationarity, $X_t \stackrel{d}{=} X_0$, so $\mathbb{E}[X_t] = \mathbb{E}[X_0] = \mu$ does not depend on t . Similarly, $(X_t, X_{t+h}) \stackrel{d}{=} (X_0, X_h)$, hence $\text{Cov}(X_t, X_{t+h}) = \text{Cov}(X_0, X_h)$ depends only on h . $\square$

Remark – The converse is false in general

A process can be weakly stationary without being strictly stationary if it has time-varying higher-order moments. For Gaussian processes (only mean and covariance matter), the two notions coincide.

3.1.2 Two canonical examples

Definition 4 – White noise

(ε_t) is a *white noise* $WN(0, \sigma^2)$ if $\mathbb{E}[\varepsilon_t] = 0$, $\text{Var}[\varepsilon_t] = \sigma^2$ and $\text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = 0$ for $h \neq 0$. A *strong white noise* requires the ε_t to be i.i.d.; a Gaussian white noise has $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$.

Example 5 – Random walk is *not* stationary

Let $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} WN(0, \sigma^2)$ and $X_t = X_{t-1} + \varepsilon_t$ with $X_0 = 0$. Then $X_t = \sum_{s=1}^t \varepsilon_s$ and

$$\mathbb{E}[X_t] = 0, \quad \text{Var}[X_t] = t\sigma^2, \quad \text{Cov}(X_t, X_{t+h}) = t\sigma^2 \text{ for } h \geq 0.$$

The variance grows linearly in t , so X_t is non-stationary. This is the canonical example used by unit-root tests in Chapter 4.

3.2 Autocovariance and autocorrelation

Definition 6 – ACVF and ACF

For a weakly stationary process with mean μ :

$$\gamma(h) = \text{Cov}(X_t, X_{t+h}), \quad \rho(h) = \frac{\gamma(h)}{\gamma(0)} \in [-1, 1].$$

Proposition 7 – Properties of γ

1. $\gamma(0) = \text{Var}[X_t] \geq 0$;
2. $|\gamma(h)| \leq \gamma(0)$ (Cauchy–Schwarz);
3. γ is even: $\gamma(-h) = \gamma(h)$;
4. γ is positive semidefinite: for any n and any $a_1, \dots, a_n \in \mathbb{R}$, $\sum_{i,j=1}^n a_i a_j \gamma(t_i - t_j) \geq 0$.

Proof. (1) and (3) follow from the definition. (2) follows from Cauchy–Schwarz applied to $X_t - \mu$ and $X_{t+h} - \mu$. (4) is the standard fact that any matrix of the form $\Sigma_{ij} = \gamma(t_i - t_j)$ is the covariance matrix of $(X_{t_1}, \dots, X_{t_n})$, hence positive semidefinite. $\square$

3.2.1 Sample ACF and Bartlett bands

Definition 8 – Sample ACVF and ACF

For $X_1, \dots, X_T$ with sample mean $\bar{X}$:

$$\hat{\gamma}(h) = \frac{1}{T} \sum_{t=1}^{T-h} (X_t - \bar{X})(X_{t+h} - \bar{X}), \quad \hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Theorem 9 – Bartlett’s formula for white noise

If $\{X_t\}$ is a white noise with finite fourth moment, then for any fixed $h \geq 1$:

$$\sqrt{T} \hat{\rho}(h) \xrightarrow{d} \mathcal{N}(0, 1).$$

The standard test bands $\pm 1.96/\sqrt{T}$ on the ACF plot are asymptotic 95% confidence intervals *under the white-noise null*.

Remark

Bartlett's formula gives a more general result for non-white processes (involves a sum over the squared autocorrelations); see Brockwell & Davis, *Time Series: Theory and Methods*, Chapter 7.

3.3 Partial autocorrelation

Definition 10 – Partial autocorrelation function

The PACF $\alpha(h)$ at lag h is the correlation between X_t and X_{t+h} after removing the linear effect of the intermediate observations $X_{t+1}, \dots, X_{t+h-1}$:

$$\alpha(h) = \text{Corr}(X_t - \hat{X}_t, X_{t+h} - \hat{X}_{t+h}),$$

where $\hat{X}_t$ and $\hat{X}_{t+h}$ are the linear projections of X_t and X_{t+h} on the span of $\{X_{t+1}, \dots, X_{t+h-1}\}$.

Remark – Recursive computation

The PACF is computed by the *Durbin–Levinson* algorithm: it solves the Yule–Walker equations of order h and reads off the coefficient of $X_{t+h-h} = X_t$ in the linear regression of X_{t+h} on $X_{t+1}, \dots, X_{t+h}$. Most software (Python `statsmodels`, R `stats`) returns the PACF in $O(h)$ operations per lag.

3.4 ACF / PACF signatures of standard processes

Process	$\rho(h)$ behaviour	$\alpha(h)$ behaviour
WN	$\rho(h) = 0$ for $h \neq 0$	$\alpha(h) = 0$ for $h \geq 1$
AR(p)	decays (exp. / sinusoidal)	cuts off at $h = p$
MA(q)	cuts off at $h = q$	decays
ARMA(p, q)	decays after $h > q$	decays after $h > p$

Box–Jenkins identification. Plot the sample ACF and PACF, then use the table above to read off candidate orders (p, q) .

3.4.1 AR(1) and MA(1) — closed-form ACF

Example 11 – AR(1) ACF

For $X_t = \phi X_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ and $|\phi| < 1$:

$$\rho(h) = \phi^{|h|}, \quad h \in \mathbb{Z}, \quad \alpha(1) = \phi, \quad \alpha(h) = 0 \text{ for } h \geq 2.$$

The ACF decays exponentially; the PACF cuts off at lag 1.

Example 12 – MA(1) ACF

For $X_t = \varepsilon_t + \theta \varepsilon_{t-1}$ with $\varepsilon_t \sim \text{WN}(0, \sigma^2)$:

$$\rho(0) = 1, \quad \rho(1) = \frac{\theta}{1 + \theta^2}, \quad \rho(h) = 0 \text{ for } h \geq 2.$$

The ACF cuts off at lag 1; the PACF decays.

Proof of $\rho(1) = \theta/(1 + \theta^2)$ in MA(1). $\text{Var}[X_t] = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})^2] = (1 + \theta^2)\sigma^2$. Then $\text{Cov}(X_t, X_{t+1}) = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})(\varepsilon_{t+1} + \theta \varepsilon_t)] = \theta \sigma^2$ since ε_t 's are uncorrelated. Hence $\rho(1) = \theta \sigma^2 / [(1 + \theta^2)\sigma^2] = \theta / (1 + \theta^2)$. $\square$

3.5 Wold decomposition

Theorem 13 – Wold’s theorem, 1938

Every *purely non-deterministic* weakly stationary process $\{X_t\}$ admits a unique representation

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \psi_0 = 1, \quad \sum_{j=0}^{\infty} \psi_j^2 < \infty,$$

where (ε_t) is a white-noise process (the *innovations*) of variance $\sigma^2 = \text{Var}[\varepsilon_t]$.

Remark – Why it matters

- **Universality:** every stationary process can be viewed as an *infinite MA process*.
- **Justification of ARMA:** ARMA models are finite-dimensional approximations of the Wold representation (with a rational $\psi(z)$).
- **Forecasting:** optimal linear predictors and their prediction errors are obtained from the Wold coefficients.

3.6 White-noise tests

After fitting a decomposition or an ARMA model, the residuals $\hat{\varepsilon}_t$ should look like white noise. Two classical tests check this jointly across several lags.

Definition 14 – Box–Pierce and Ljung–Box statistics

For sample ACF $\hat{\rho}(h)$ of the residuals up to lag m :

$$Q_{\text{BP}} = T \sum_{h=1}^m \hat{\rho}(h)^2, \quad Q_{\text{LB}} = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}(h)^2}{T-h}.$$

Theorem 15 – Asymptotic distribution under H_0

Under H_0 : *the residuals are white noise*, both statistics are asymptotically χ_{m-p-q}^2 , where $p+q$ is the number of fitted ARMA parameters (zero if no parametric model has been fitted).

Decision rule. Reject H_0 at level α if $Q > \chi_{m-p-q, 1-\alpha}^2$, equivalently $p\text{-value} < \alpha$.

Choosing m . A few rules of thumb:

- For yearly data: $m \in \{10, 20\}$.
- For monthly data: $m \in \{12, 24, 36\}$ (covers seasonality).
- For high-frequency data: m proportional to expected autocorrelation length.

Warning

The Ljung–Box test is preferred over Box–Pierce in finite samples: its small-sample distribution is closer to the χ^2 asymptotic limit, especially when m/T is not negligible.

Summary

Key takeaways.

- *Strict stationarity* = invariance of all finite-dimensional distributions; *weak stationarity* = only first two moments. Strict $\Rightarrow$ weak (under L^2); converse holds for Gaussian processes.

- The *autocovariance* $\gamma(h)$ describes the dependence structure; the *autocorrelation* $\rho(h)$ is its normalised version.
- ACF and PACF have characteristic patterns that allow *Box-Jenkins identification* of AR(p) / MA(q) / ARMA(p,q).
- *Wold decomposition*: every stationary process is an infinite moving average of innovations.
- *Ljung-Box test* validates that residuals look like white noise.

Chapter 4

Non-stationary time series

Introduction

Most real-world time series are *not* stationary: GDP grows, prices drift, populations expand, atmospheric CO₂ accumulates. Applying the tools of Chapter 3 (ACF, ARMA) to a raw non-stationary series gives misleading results — the famous *spurious regression* phenomenon. This chapter formalises non-stationarity, introduces the algebraic operators (backshift, difference) and the unit-root tests (ADF, KPSS) needed to detect it, and culminates in the *ARIMA* and *SARIMA* families that combine differencing with the ARMA structure of Chapter 5.

4.1 Limits of the stationarity assumption

Warning – Spurious regression

If X_t and Y_t are two *independent* random walks and one fits the linear regression $Y_t = \beta_0 + \beta_1 X_t + u_t$ by OLS:

- the standard t -statistic for $\hat{\beta}_1$ rejects the null $\beta_1 = 0$ at any level with probability tending to *one* as the sample size grows;
- the R^2 stays large (often > 0.5);

even though the two series are *independent*. Granger & Newbold (1974) called this the *spurious regression* effect; it is caused by the non-stationarity of X and Y .

Remark – Practical consequence

Before applying any classical inference (regression, ARMA, t -test), *check stationarity*. If the series is non-stationary, transform it first (typically by differencing).

4.1.1 Trend stationary vs. difference stationary

Definition 1 – TS and DS processes

- *Trend stationary* (TS): $X_t = f(t) + \varepsilon_t$ where f is deterministic and ε_t is stationary. Shocks have *transient* effects.
- *Difference stationary* (DS): ΔX_t is stationary, but X_t is not. The series has a *unit root*; shocks have *permanent* effects.

Remark

TS series are made stationary by *detrending*; DS series, by *differencing*. Visual inspection alone cannot reliably distinguish the two — a slowly trending DS process looks identical to a TS one.

We use *unit-root tests* (Section 4.3).

4.2 Backshift and difference operators

Definition 2 – Backshift operator

$BX_t = X_{t-1}$, and $B^k X_t = X_{t-k}$. For polynomials $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$, the AR(p) and MA(q) processes are written $\phi(B)X_t = \varepsilon_t$ and $X_t = \theta(B)\varepsilon_t$.

Proposition 3 – Stationarity through the AR polynomial

The AR(p) process $\phi(B)X_t = \varepsilon_t$ is (weakly) stationary if and only if every root of $\phi(z) = 0$ lies *outside* the closed unit disk.

Example 4 – AR(1)

$\phi(z) = 1 - \phi z$ has root $z = 1/\phi$. $|1/\phi| > 1 \Leftrightarrow |\phi| < 1$. We recover the familiar AR(1) stationarity condition.

Definition 5 – Difference operator

$\Delta = 1 - B$, so $\Delta X_t = X_t - X_{t-1}$. $\Delta^d = (1 - B)^d$ is the d -th difference. The seasonal difference of period s is $\Delta_s = 1 - B^s$.

Example 6 – Higher-order differencing

$\Delta^2 X_t = (1 - B)^2 X_t = X_t - 2X_{t-1} + X_{t-2}$. A quadratic trend $X_t = a + bt + ct^2$ is reduced to a constant by Δ^2 : $\Delta^2 X_t = 2c$, which is stationary.

Definition 7 – Integration order $I(d)$

$X_t \sim I(d)$ if $\Delta^d X_t$ is stationary but $\Delta^{d-1} X_t$ is not. White noise is $I(0)$; a random walk is $I(1)$; a random walk with a stochastic trend is often $I(2)$.

4.3 Unit-root tests

4.3.1 Augmented Dickey–Fuller (ADF) test

Definition 8 – ADF regression

Starting from an AR(1) $X_t = \rho X_{t-1} + \varepsilon_t$, substitute $\rho = 1 + \gamma$ and subtract X_{t-1} :

$$\Delta X_t = \gamma X_{t-1} + \varepsilon_t,$$

augmented with p lags of ΔX_t to whiten the error:

$$\Delta X_t = \gamma X_{t-1} + \sum_{i=1}^p \alpha_i \Delta X_{t-i} + \varepsilon_t.$$

Theorem 9 – ADF test

- $H_0 : \gamma = 0$ — unit root, X_t is non-stationary.
- $H_1 : \gamma < 0$ — X_t is stationary.

The t -statistic of $\hat{\gamma}$ *does not* follow a Student's t under H_0 ; its asymptotic distribution is the *Dickey–Fuller distribution*, with 5% critical value ≈ -2.86 for the constant-only specification (and ≈ -3.41 with constant and trend).

Remark – Three flavours

- *No constant, no trend* (rare): X_t should fluctuate around zero.
- *Constant only* (most common): X_t should fluctuate around a non-zero level.
- *Constant and trend*: X_t has a deterministic linear trend.

The choice affects the critical values; the Python function `adfuller` accepts a `regression` argument.

4.3.2 KPSS test

Definition 10 – KPSS test, Kwiatkowski et al. 1992

H_0 : stationarity (around a level or a trend), H_1 : unit-root non-stationarity.

The test statistic is built from the partial sums of OLS residuals versus a long-run variance estimator. 5% critical values: 0.463 for level stationarity, 0.146 for trend stationarity. Reject H_0 if the statistic *exceeds* the critical value.

Remark – Combining ADF and KPSS

The two tests have *opposite* null hypotheses, which clarifies ambiguous cases:

	ADF rejects	ADF does not reject
KPSS rejects	contradiction	non-stationary
KPSS does not reject	stationary	inconclusive (small T)

The contradiction case typically arises when there is a structural break.

4.3.3 Phillips–Perron test (mention)

Remark

The *Phillips–Perron* (PP) test is an alternative to ADF that adjusts the test statistic for heteroscedasticity and autocorrelation non-parametrically (instead of via lagged differences). Same null hypothesis as ADF, similar critical values.

4.4 ARIMA(p, d, q)

Definition 11 – ARIMA model

$X_t \sim \text{ARIMA}(p, d, q)$ if its d -th difference is a stationary $\text{ARMA}(p, q)$:

$$\phi(B)(1-B)^d X_t = \theta(B)\varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2).$$

Remark – Practical recipe

1. Plot X_t ; identify visual non-stationarity.
2. Test (ADF + KPSS); difference until stationary; choose d .
3. ACF/PACF of $\Delta^d X_t \Rightarrow$ candidate (p, q) .
4. MLE estimate of the $\text{ARMA}(p, q)$ coefficients.

5. Ljung–Box on residuals; if not white, increase p or q .
6. Forecast, integrate back.

Example 12 – ARIMA(1, 1, 1)

$\phi(B) = 1 - \phi B$, $\theta(B) = 1 + \theta B$, $d = 1$:

$$(1 - \phi B)(1 - B)X_t = (1 + \theta B)\varepsilon_t.$$

With $W_t = \Delta X_t$: $W_t = \phi W_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$, i.e. $W_t \sim \text{ARMA}(1, 1)$. Starting from a known initial value X_0 , integrate back: $X_t = X_0 + \sum_{s=1}^t W_s$.

4.5 Seasonal ARIMA (SARIMA)

When the series exhibits seasonality of period s , classical differencing $(1 - B)^d$ alone is not enough — we also apply seasonal differencing $(1 - B^s)^D$ and seasonal ARMA polynomials.

Definition 13 – SARIMA(p, d, q)(P, D, Q)_s

$$\phi(B) \Phi(B^s) (1 - B)^d (1 - B^s)^D X_t = \theta(B) \Theta(B^s) \varepsilon_t,$$

where $\Phi(z) = 1 - \Phi_1 z - \dots - \Phi_P z^P$ and $\Theta(z) = 1 + \Theta_1 z + \dots + \Theta_Q z^Q$ are the *seasonal* AR and MA polynomials.

Example 14 – SARIMA(1, 1, 1)(1, 1, 1)₁₂

$$(1 - \phi B)(1 - \Phi B^{12})(1 - B)(1 - B^{12})X_t = (1 + \theta B)(1 + \Theta B^{12})\varepsilon_t.$$

This model is the workhorse for monthly time series with both short-term persistence and yearly seasonality (sales, climate, energy demand).

Remark – Identification

After the differencing step, look at the ACF/PACF of $(1 - B)^d(1 - B^s)^D X_t$:

- significant spikes at lags $s, 2s, 3s, \dots$ in the ACF $\Rightarrow$ seasonal MA component (order Q);
- spikes at $s, 2s, \dots$ in the PACF $\Rightarrow$ seasonal AR component (order P);
- spikes at small lags $\Rightarrow$ ordinary p, q .

Summary

Key takeaways.

- Most real series are non-stationary; *spurious regression* can fool naive applications of OLS or ARMA.
- Distinguish *trend stationary* (detrend) from *difference stationary* (difference) using the ADF and KPSS tests.
- The backshift B and difference Δ operators give a compact algebra for non-stationary models; integration order $I(d)$ counts the number of differencings needed.
- $\text{ARIMA}(p, d, q) = \text{ARMA}$ on the d -th difference; *SARIMA* adds seasonal AR, MA and differencing.
- In Python, `statsmodels.tsa.stattools.adfuller` and `kpss`, plus **SARIMAX**, cover the whole workflow.

Chapter 5

ARMA models

Introduction

Chapter 3 introduced stationarity and the sample ACF/PACF; Chapter 4 showed how to turn a non-stationary series into a stationary one by differencing. This chapter assembles the central modelling family for stationary time series: the *autoregressive moving-average* processes, $\text{ARMA}(p, q)$. We give their precise definitions, state the stationarity and invertibility conditions in terms of polynomial roots, derive their ACF/PACF *signatures*, and detail the practical workflow — the Box–Jenkins method — for identification, estimation, diagnostics and forecasting.

5.1 Autoregressive process $\text{AR}(p)$

Definition 1 – Autoregressive process $\text{AR}(p)$

A process $(X_t)_{t \in \mathbb{Z}}$ is autoregressive of order p if

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2),$$

with $\phi_p \neq 0$. Equivalently, with $\Phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$,

$$\Phi(B) X_t = \varepsilon_t.$$

Proposition 2 – Stationarity of $\text{AR}(p)$

Equation $\Phi(B) X_t = \varepsilon_t$ admits a unique causal stationary solution if and only if every root z_i of $\Phi(z) = 0$ satisfies $|z_i| > 1$. In that case

$$X_t = \Phi^{-1}(B) \varepsilon_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j},$$

with $\sum_j |\psi_j| < \infty$.

Example 3 – $\text{AR}(1)$ moments

For $X_t = \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$, iterating gives $X_t = \sum_{j \geq 0} \phi^j \varepsilon_{t-j}$, hence

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \frac{\sigma^2}{1 - \phi^2}, \quad \rho(h) = \phi^{|h|}.$$

The ACF **decays geometrically** and the PACF **cuts off** after lag 1 ($\alpha(1) = \phi$, $\alpha(h) = 0$ for $h \geq 2$).

Proposition 4 – Yule–Walker equations

Let $X_t \sim \text{AR}(p)$ be stationary. Multiplying the equation by X_{t-h} and taking expectations gives, for $h = 1, \dots, p$,

$$\rho(h) = \phi_1 \rho(h-1) + \phi_2 \rho(h-2) + \dots + \phi_p \rho(h-p).$$

In matrix form $\mathbf{R}\boldsymbol{\phi} = \mathbf{r}$, the system is invertible under the stationarity condition; replacing ρ by the sample ACF $\hat{\rho}$ gives the Yule–Walker estimator $\hat{\phi}_{\text{YW}}$.

Example 5 – AR(2) stationarity triangle

For $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$, the roots of $1 - \phi_1 z - \phi_2 z^2 = 0$ lie outside the unit disk iff

$$\phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1, \quad -1 < \phi_2 < 1.$$

This is the Schur stability condition for the AR(2) polynomial.

5.2 Moving-average process MA(q)

Definition 6 – Moving-average process MA(q)

A process $(X_t)_{t \in \mathbb{Z}}$ is moving-average of order q if

$$X_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2),$$

with $\theta_q \neq 0$. Equivalently, $X_t = \Theta(B) \varepsilon_t$ with $\Theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$.

Proposition 7 – Moments and ACF of MA(q)

Every MA(q) is (weakly) stationary, with

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \sigma^2(1 + \theta_1^2 + \dots + \theta_q^2),$$

and autocovariance

$$\gamma(h) = \begin{cases} \sigma^2(\theta_h + \theta_1 \theta_{h+1} + \dots + \theta_{q-h} \theta_q) & 1 \leq h \leq q, \\ 0 & h > q, \end{cases}$$

with $\theta_0 = 1$. In particular $\rho(h) = 0$ for $h > q$: the ACF **cuts off** at lag q .

Definition 8 – Invertibility

An MA(q) is *invertible* if it admits an AR(∞) representation

$$\varepsilon_t = X_t - \pi_1 X_{t-1} - \pi_2 X_{t-2} - \dots, \quad \sum_j |\pi_j| < \infty.$$

This holds if and only if every root of $\Theta(z) = 0$ lies *outside* the closed unit disk.

Remark – Why invertibility matters

Without invertibility, two parameter sets give the same ACF and so are indistinguishable from the data. The MA(1) models $X_t = \varepsilon_t + \theta \varepsilon_{t-1}$ and $X_t = \tilde{\varepsilon}_t + (1/\theta) \tilde{\varepsilon}_{t-1}$ have identical autocovariance functions; we conventionally choose the invertible one, $|\theta| < 1$.

5.3 ARMA(p, q)

Definition 9 – ARMA(p, q)

A process (X_t) is ARMA(p, q) if it satisfies

$$\Phi(B) X_t = \Theta(B) \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2),$$

with $\Phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$, $\Theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$, and Φ, Θ sharing no common root.

Proposition 10 – Stationarity, invertibility and existence

The equation $\Phi(B)X_t = \Theta(B)\varepsilon_t$ admits a unique causal stationary invertible solution if and only if

$$|z_i| > 1 \text{ for every root of } \Phi, \quad |z_j| > 1 \text{ for every root of } \Theta.$$

The solution is then

$$X_t = \Phi^{-1}(B) \Theta(B) \varepsilon_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}.$$

Remark

The MA(∞) coefficients (ψ_j) are obtained by polynomial division of $\Theta(z)$ by $\Phi(z)$ around $z = 0$. They control both the autocovariance and the forecast variance (Section 5.8).

5.4 Box–Jenkins identification

The choice of (p, q) is guided by the sample ACF and PACF, whose *signatures* differ across model classes:

Model	ACF $\rho(h)$	PACF $\alpha(h)$
AR(p)	exponential decay	cuts off after lag p
MA(q)	cuts off after lag q	exponential decay
ARMA(p, q)	exponential decay (after q)	exponential decay (after p)

Remark – Bartlett bands

“Cuts off” is read as $\hat{\rho}(h)$ falling *inside* the $\pm 1.96/\sqrt{T}$ band for h larger than the cutoff lag. “Decays” means the values stay outside the band but shrink in absolute value.

Example 11 – Reading a sample ACF/PACF

On a series of length $T = 200$, the sample PACF shows $\hat{\alpha}(1) = 0.62$, $\hat{\alpha}(2) = -0.04$, $\hat{\alpha}(3) = 0.02$ and the sample ACF decays geometrically. The Bartlett band is $\pm 1.96/\sqrt{200} \approx \pm 0.139$, so $\hat{\alpha}(h)$ is inside the band for $h \geq 2$: the PACF cuts off at lag 1, suggesting AR(1).

5.5 Estimation

5.5.1 Yule–Walker (AR only)

For pure AR(p), plug $\hat{\rho}$ into Proposition 5.1 to obtain the Yule–Walker estimate. This is method-of-moments: *consistent* but not asymptotically efficient.

5.5.2 Conditional sum of squares (CSS)

Conditioning on the first p observations and assuming $\varepsilon_t = 0$ for $t \leq 0$, the CSS estimator is

$$\hat{\theta}_{\text{CSS}} = \arg \min_{\theta} \sum_{t=p+1}^T \hat{\varepsilon}_t(\theta)^2,$$

where $\hat{\varepsilon}_t$ is computed recursively from the ARMA equation. CSS is fast and equivalent to MLE in large samples under Gaussian innovations.

5.5.3 Maximum likelihood (MLE)

Under $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, the joint distribution of $(X_1, \dots, X_T)$ is multivariate normal with covariance matrix $\Gamma_T(\theta)$. The log-likelihood

$$\log \mathcal{L}(\theta, \sigma^2) = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \log \det \Gamma_T(\theta) - \frac{1}{2} \mathbf{X}^\top \Gamma_T(\theta)^{-1} \mathbf{X}$$

is maximised numerically. The standard implementation in `statsmodels` uses a state-space representation and the Kalman filter, which scale linearly in T and provide asymptotic standard errors.

Theorem 12 – Asymptotic distribution of the MLE

Under regularity assumptions (stationarity, invertibility, no common roots), as $T \rightarrow \infty$,

$$\sqrt{T}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}),$$

where $I(\theta_0)$ is the Fisher information matrix evaluated at the true parameter θ_0 .

5.6 Model selection

Definition 13 – Information criteria

For a fitted ARMA(p, q) with $k = p + q + 1$ free parameters (including σ^2),

$$\text{AIC} = -2 \log \hat{\mathcal{L}} + 2k, \quad \text{BIC} = -2 \log \hat{\mathcal{L}} + k \log T.$$

The chosen order minimises the criterion.

Remark – AIC vs BIC

AIC favours predictive accuracy and tends to over-fit on small samples; BIC penalises complexity more and is consistent for the true order under standard assumptions. When the two disagree, prefer the BIC choice for parsimony and confirm with the residual diagnostics.

5.7 Residual diagnostics

Remark – Adequacy checks

A well-specified ARMA model produces residuals $\hat{\varepsilon}_t = X_t - \hat{X}_{t|t-1}$ that are *indistinguishable from white noise*. Standard checks:

- time plot of $\hat{\varepsilon}_t$ – mean stable around zero, variance constant;
- sample ACF of $\hat{\varepsilon}_t$ – spikes inside Bartlett bands;
- Ljung–Box test;
- QQ-plot vs $\mathcal{N}(0, 1)$ for normality (Jarque–Bera test).

Theorem 14 – Ljung–Box test on residuals

$$Q_{\text{LB}}(m) = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}_{\varepsilon}(h)^2}{T-h}.$$

Under the null that $\hat{\varepsilon}_t$ is white noise, after fitting an ARMA(p, q),

$$Q_{\text{LB}}(m) \xrightarrow{d} \chi_{m-p-q}^2.$$

Reject the null if $Q_{\text{LB}}(m)$ exceeds the $\chi_{m-p-q, 1-\alpha}^2$ quantile.

5.8 Forecasting

Definition 15 – Best linear forecast

The minimum mean-squared-error linear forecast at horizon h , given the past $\mathcal{F}_T = \sigma(X_T, X_{T-1}, \dots)$, is

$$\hat{X}_{T+h|T} = \mathbb{E}[X_{T+h} | \mathcal{F}_T].$$

The forecast error is $e_{T+h} = X_{T+h} - \hat{X}_{T+h|T}$.

Proposition 16 – Forecast error variance

With the MA(∞) representation $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$,

$$e_{T+h} = \sum_{j=0}^{h-1} \psi_j \varepsilon_{T+h-j},$$

hence

$$\mathbb{E}[e_{T+h}] = 0, \quad \text{Var}(e_{T+h}) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

Example 17 – Forecast for AR(1)

With $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$:

$$\hat{X}_{T+h|T} = \phi^h X_T, \quad \text{Var}(e_{T+h}) = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2}.$$

As $h \rightarrow \infty$, $\hat{X}_{T+h|T} \rightarrow 0$ (the unconditional mean) and the variance tends to $\sigma^2/(1 - \phi^2)$ (the unconditional variance).

Remark – Confidence intervals

Under Gaussian innovations, the $1 - \alpha$ confidence interval is

$$\hat{X}_{T+h|T} \pm z_{1-\alpha/2} \sigma \sqrt{\sum_{j=0}^{h-1} \psi_j^2}.$$

For stationary ARMA, the width converges to a finite limit; for ARIMA (Chapter 4), it grows without bound.

Summary

Key takeaways.

- $AR(p)$ depends on past values; $MA(q)$ depends on past shocks; $ARMA(p, q)$ combines both.
- Stationarity = roots of Φ outside the unit disk; invertibility = roots of Θ outside the unit disk.
- ACF/PACF signatures: AR cuts the PACF, MA cuts the ACF, ARMA shows double exponential decay.
- Estimation: Yule–Walker for pure AR; CSS or MLE for general ARMA. `statsmodels.tsa.arima.ARIMA` provides both.
- Model selection by AIC / BIC; validate with Ljung–Box on residuals.
- Forecasts widen with horizon according to $\text{Var}(e_{T+h}) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$.

Chapter 6

ARCH and GARCH models

Introduction

The ARMA family of Chapter 5 captures the dynamics of the *conditional mean*, but assumes a constant variance. On financial returns, on macroeconomic surprises and on many physical measurements, this assumption is violated: large absolute deviations cluster in time, and the variance itself is predictable. This chapter introduces the *autoregressive conditional heteroskedasticity* (ARCH) family that models the conditional variance directly, generalises it to GARCH(p, q), derives the stationarity conditions, presents the maximum-likelihood estimator, and applies the model to volatility forecasting and Value-at-Risk.

6.1 Stylised facts of financial returns

Remark – Empirical regularities

For daily log-returns $r_t = \log(P_t/P_{t-1})$ on broad equity indices, currencies and commodities:

1. $\mathbb{E}[r_t] \approx 0$ (small drift).
2. ACF of r_t : nearly flat – returns are approximately serially uncorrelated.
3. ACF of r_t^2 (or $|r_t|$): *strongly* positive at many lags – volatility is autocorrelated.
4. Marginal kurtosis $\hat{\kappa} > 3$: heavier tails than Gaussian.
5. Leverage effect: negative shocks raise future volatility more than positive shocks of the same size.

Remark – Conditional vs. unconditional

Let $\mathcal{F}_{t-1} = \sigma(r_{t-1}, r_{t-2}, \dots)$ be the information set up to $t-1$.

- *Unconditional variance*: $\sigma^2 = \text{Var}(r_t)$, the long-run, average risk.
- *Conditional variance*: $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$, the short-run risk knowing the past.

ARCH and GARCH models *specify* σ_t^2 while keeping σ^2 finite – the process is unconditionally homoskedastic but conditionally heteroskedastic.

Definition 1 – Mean–variance decomposition

A second-order martingale-difference innovation $\varepsilon_t = r_t - \mu_t$ is written

$$\varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{D}(0, 1), \quad \mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}] = 0, \quad \text{Var}(\varepsilon_t | \mathcal{F}_{t-1}) = \sigma_t^2.$$

$\mathcal{D}$ is the *innovation distribution*, typically $\mathcal{N}(0, 1)$ or Student- t with low degrees of freedom.

6.2 ARCH(q) model

Definition 2 – ARCH(q), Engle 1982

The *autoregressive conditional heteroskedasticity* model of order q specifies

$$\varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2,$$

with $z_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $\alpha_0 > 0$, and $\alpha_i \geq 0$ for $i = 1, \dots, q$.

Proposition 3 – Properties of ARCH(q)

Assume the covariance-stationarity condition $\sum_{i=1}^q \alpha_i < 1$. Then:

1. $\mathbb{E}[\varepsilon_t] = 0$ and (ε_t) is a white noise (uncorrelated in time, but not independent).
2. $\mathbb{E}[\varepsilon_t^2] = \alpha_0 / (1 - \sum \alpha_i) =: \sigma^2$.
3. (ε_t^2) admits an AR(q) representation $\varepsilon_t^2 = \alpha_0 + \sum \alpha_i \varepsilon_{t-i}^2 + \nu_t$, with $\nu_t = \varepsilon_t^2 - \sigma_t^2$ a martingale difference.
4. Excess kurtosis: under Gaussian innovations ARCH(1) has kurtosis $3(1 - \alpha_1^2)/(1 - 3\alpha_1^2) > 3$ when $\alpha_1^2 < 1/3$.

Remark – Why squares?

Squaring past residuals makes the response *sign-blind*: a -3σ surprise and a $+3\sigma$ surprise both raise σ_{t+1}^2 . To allow asymmetric responses (leverage), one moves to GJR-GARCH or EGARCH (Section 6.7).

6.3 GARCH(p, q) model

In practice, ARCH(q) requires many lags q to capture the slow decay of squared-return autocorrelations, leading to long, poorly identified parameter vectors. Bollerslev (1986) generalised the model by including past conditional variances themselves.

Definition 4 – GARCH(p, q), Bollerslev 1986

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2,$$

with $\alpha_0 > 0$, $\alpha_i, \beta_j \geq 0$.

Proposition 5 – Stationarity of GARCH(p, q)

(ε_t) admits a covariance-stationary solution with finite unconditional variance if and only if

$$\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j < 1.$$

The unconditional variance is then

$$\sigma^2 = \mathbb{E}[\varepsilon_t^2] = \frac{\alpha_0}{1 - \sum_i \alpha_i - \sum_j \beta_j}.$$

Example 6 – GARCH(1,1): the workhorse

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

On daily equity returns, typical estimates are $\alpha_1 \approx 0.05$ – 0.10 , $\beta_1 \approx 0.85$ – 0.92 , $\alpha_1 + \beta_1 \approx 0.95$ – 0.99 . Volatility is *very persistent* but still mean-reverting.

Remark – IGARCH and persistence

The boundary case $\alpha_1 + \beta_1 = 1$ defines an *IGARCH*: shocks have a permanent effect on the conditional variance. The unconditional variance is then infinite (the model is strictly stationary but not covariance-stationary). Estimated $\alpha_1 + \beta_1$ very close to 1 is a sign that the assumed GARCH(1,1) may be missing a structural break.

6.4 Maximum-likelihood estimation

Definition 7 – Conditional Gaussian likelihood

Assuming $z_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ and $r_t = \mu + \varepsilon_t$, the conditional log-likelihood is

$$\log \mathcal{L}(\boldsymbol{\theta}) = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T \left(\log \sigma_t^2(\boldsymbol{\theta}) + \frac{(r_t - \mu)^2}{\sigma_t^2(\boldsymbol{\theta})} \right),$$

with σ_t^2 computed recursively from the GARCH equation. $\boldsymbol{\theta} = (\mu, \alpha_0, \alpha_1, \dots, \alpha_q, \beta_1, \dots, \beta_p)$.

Theorem 8 – Asymptotic properties of the QMLE

Under regularity conditions (positivity, stationarity, bounded fourth moment), the Gaussian quasi-MLE $\hat{\boldsymbol{\theta}}_{\text{QMLE}}$ is consistent and asymptotically normal,

$$\sqrt{T}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(0, \Sigma),$$

even if the true innovation distribution is non-Gaussian (e.g. Student- t). The asymptotic variance Σ takes a sandwich form $\Sigma = J^{-1} I J^{-1}$.

Remark – Practical numerics

- Returns are typically rescaled (multiplied by 100 to work in percentage units) for numerical stability.
- Initial σ_1^2 is set to the sample variance, or to the unconditional value $\alpha_0/(1 - \alpha_1 - \beta_1)$ at the current parameters.
- In Python, the `arch` package (`arch_model`) does the rescaling, parameter constraints and likelihood maximisation automatically.

6.5 Diagnostics

6.5.1 Engle's LM test for ARCH effects

Definition 9 – Engle's LM test

Apply on the residual $\hat{\varepsilon}_t$ of a mean model (constant or ARMA). Run the auxiliary regression

$$\hat{\varepsilon}_t^2 = \gamma_0 + \gamma_1 \hat{\varepsilon}_{t-1}^2 + \dots + \gamma_q \hat{\varepsilon}_{t-q}^2 + u_t,$$

get its R^2 . Test

$$H_0 : \gamma_1 = \dots = \gamma_q = 0, \quad LM = T \cdot R^2 \xrightarrow{d} \chi_q^2.$$

Reject H_0 at level α if $LM > \chi_{q,1-\alpha}^2$.

Remark – How to use it

Run Engle's LM *before* fitting GARCH (to confirm there is an ARCH effect to model) and *after* (on the standardised residuals $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$, to confirm the model has captured it).

6.5.2 Standardised residual diagnostics

Remark

A well-fitted GARCH produces standardised residuals that look like white noise both in level and in square:

- ACF of $\hat{z}_t$ inside Bartlett bands;
- ACF of $\hat{z}_t^2$ inside Bartlett bands;
- Engle LM on $\hat{z}_t^2$ does not reject;
- marginal distribution close to $\mathcal{D}(0, 1)$ (QQ-plot, Kolmogorov–Smirnov).

Persistent failures in $\hat{z}_t^2$ suggest a higher-order GARCH or a Student- t innovation; failures only in $\hat{z}_t^-$ suggest a leverage extension (GJR, EGARCH).

6.6 Volatility forecasting and Value-at-Risk

Proposition 10 – GARCH(1, 1) multi-step forecast

Let $\hat{\sigma}_T^2$ and $\hat{\varepsilon}_T$ be the values obtained at the end of the sample. Then

$$\hat{\sigma}_{T+1}^2 = \alpha_0 + \alpha_1 \hat{\varepsilon}_T^2 + \beta_1 \hat{\sigma}_T^2,$$

and for $h \geq 2$,

$$\hat{\sigma}_{T+h}^2 = \sigma^2 + (\alpha_1 + \beta_1)^{h-1} (\hat{\sigma}_{T+1}^2 - \sigma^2),$$

where $\sigma^2 = \alpha_0 / (1 - \alpha_1 - \beta_1)$ is the unconditional variance. The forecast *mean-reverts* to σ^2 at rate $\alpha_1 + \beta_1$.

Definition 11 – Conditional Value-at-Risk

At horizon 1 and confidence $1 - \alpha$ (typically $\alpha = 1\%$ or 5%), assuming Gaussian innovations,

$$\text{VaR}_{T+1}(\alpha) = -(\hat{\mu} + z_\alpha \hat{\sigma}_{T+1}),$$

with $z_\alpha = \Phi^{-1}(\alpha)$ ($z_{0.01} = -2.326$, $z_{0.05} = -1.645$). The interpretation: with probability $1 - \alpha$, the loss does not exceed $\text{VaR}_{T+1}(\alpha)$.

Remark – Static vs dynamic VaR

The classical static VaR uses the *sample* standard deviation $\hat{\sigma}$. The GARCH-VaR uses $\hat{\sigma}_{T+1}$. In calm regimes, $\hat{\sigma}_{T+1} < \hat{\sigma}$ and the dynamic VaR is tighter; in turbulent regimes (just after a market shock), $\hat{\sigma}_{T+1} > \hat{\sigma}$ and the dynamic VaR is wider, accurately reflecting the elevated short-term risk.

6.7 Extensions (overview)

Remark – EGARCH (Nelson 1991)

The exponential GARCH models $\log \sigma_t^2$, removing the positivity constraints on the parameters and naturally allowing asymmetric responses:

$$\log \sigma_t^2 = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \log \sigma_{t-1}^2.$$

$\gamma < 0$ is the typical sign of the leverage effect on equity returns.

Remark – GJR-GARCH (Glosten–Jagannathan–Runkle 1993)

Adds a threshold term that activates only on negative shocks:

$$\sigma_t^2 = \alpha_0 + (\alpha_1 + \gamma \mathbb{1}_{\varepsilon_{t-1} < 0}) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

$\gamma > 0$ implies that bad news raises future volatility more than good news of the same magnitude.

Summary

Key takeaways.

- Financial returns are uncorrelated but their squares are autocorrelated – volatility comes in clusters.
- ARCH(q): σ_t^2 is a linear function of past squared shocks. GARCH(p, q) adds memory through past conditional variances.
- GARCH(1, 1) is stationary iff $\alpha_1 + \beta_1 < 1$, with unconditional variance $\alpha_0 / (1 - \alpha_1 - \beta_1)$.
- Engle's LM test detects ARCH effects in residuals; standardised residuals validate the fit.
- Conditional volatility forecasts mean-revert to the unconditional variance at rate $\alpha_1 + \beta_1$; Value-at-Risk uses these forecasts directly.
- Extensions: EGARCH and GJR-GARCH for asymmetric (leverage) effects.

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